

SEARAN KUGANESAN

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EDUCATION

Western University

London, Canada

B.E.Sc. Software Engineering

Graduated: 06/2026

- **Relevant coursework:** Machine learning, Databases, Data Structures, Web Development, Algorithm Design and Analysis, Computer Architecture, Operating Systems, Computational Complexity, Artificial Intelligence

EXPERIENCE

Mitsubishi Heavy Industries

05/2024 – 08/2025

Software Engineering Intern

Mississauga, Canada

- Engineered full-stack component tracking system leveraging **React.js** frontend and **Python/Flask RESTful API** backend serving **2,000+** aircraft across **100+ operators** with interactive **D3.js/Chart.js** data visualizations
- Engineered fleet prediction platform processing **400,000+ monthly records** from **SQL Server/Oracle** databases using **pandas/NumPy ETL pipelines** with automated exception handling and validation
- Implemented **machine learning** trend analysis with scikit-learn **regression models** for utilization forecasting across **50+ operators** and **6 regional markets** with **statistical validation**
- Generated automated monthly aircraft utilization forecasts and reliability reports through **Python data processing pipelines** with scheduled task automation, reducing manual processing time by 85%
- Designed **Oracle database architecture** with optimized indexing and query performance tuning and built **scheduling optimization engine** coordinating maintenance workflows across **2,000+ entities**
- Developed comprehensive unit and integration test suites using **pytest** and **Jest**, achieving 85% code coverage and implementing **CI/CD pipelines** with **automated testing** to ensure reliability across production deployments
- Created comprehensive technical documentation and **API specifications**, and mentored junior interns

PROJECTS

CraftTraq – Multi-Tenant SaaS Platform for Trade Contractors | crafttraq.com | *React 19, TypeScript, FastAPI, PostgreSQL*

- Built end-to-end job lifecycle management, from **shareable client quotes** that auto-convert into jobs to **drag-and-drop crew scheduling** and status-tracked job records, live in production with tiered **Stripe** billing
- Engineered scheduling with **immutable task-assignment snapshots** reconciled against live crew membership
- Built **invoicing, time tracking, and inventory management** across **28+ tables** with a **PDF generation pipeline**
- Integrated **Stripe, QuickBooks OAuth2** payroll sync, **Twilio SMS**, and **Supabase Realtime** for live job-status updates across the **Progressive Web App**

Portfolio Risk Dashboard | *React, Python, Flask, MongoDB, NumPy, Pandas, SciPy*

- Built full-stack financial analytics application using **React/Vite** and **Flask** for comprehensive Modern Portfolio Theory analysis and Value at **Risk calculations** with interactive dashboards
- Engineered **Python backend** with **yfinance API** integration and **NumPy/Pandas/SciPy** libraries for real-time market data visualization using **recharts** with **data processing**
- Implemented statistical portfolio optimization algorithms for efficient frontier analysis and **risk metrics**

Multi-Model Text Classification Pipeline | *Python, TensorFlow, BERT, scikit-learn, PostgreSQL, MLflow*

- Built ensemble system combining **BERT embeddings** with CNN and **BiLSTM architectures** for categorization
- Designed **experiment tracking** with **MLflow logging** model configurations and **hyperparameter combinations**
- Created **data augmentation pipeline** using **back-translation** and synonym replacement to expand training set

TECHNICAL SKILLS

Languages: Python, Java, C/C++, Kotlin, C#, TypeScript, SQL, MATLAB, HTML/CSS, Bash

Frameworks: React, Node, Express, Spring Boot, Django, .NET, Flask, FastAPI, Vite, Tailwind CSS

Technologies/Tools: Git, Docker, MySQL, MongoDB, PostgreSQL, Supabase, AWS (S3, Lambda, Transcribe), Google Cloud Platform, Cloudflare, Stripe, Twilio, Selenium, Postman, Jira, Articulate Storyline, Excel, Figma

Production & Infrastructure: CI/CD, Unit-testing, Stress/Exhaustion Testing, Regression Testing, Git, Azure DevOps